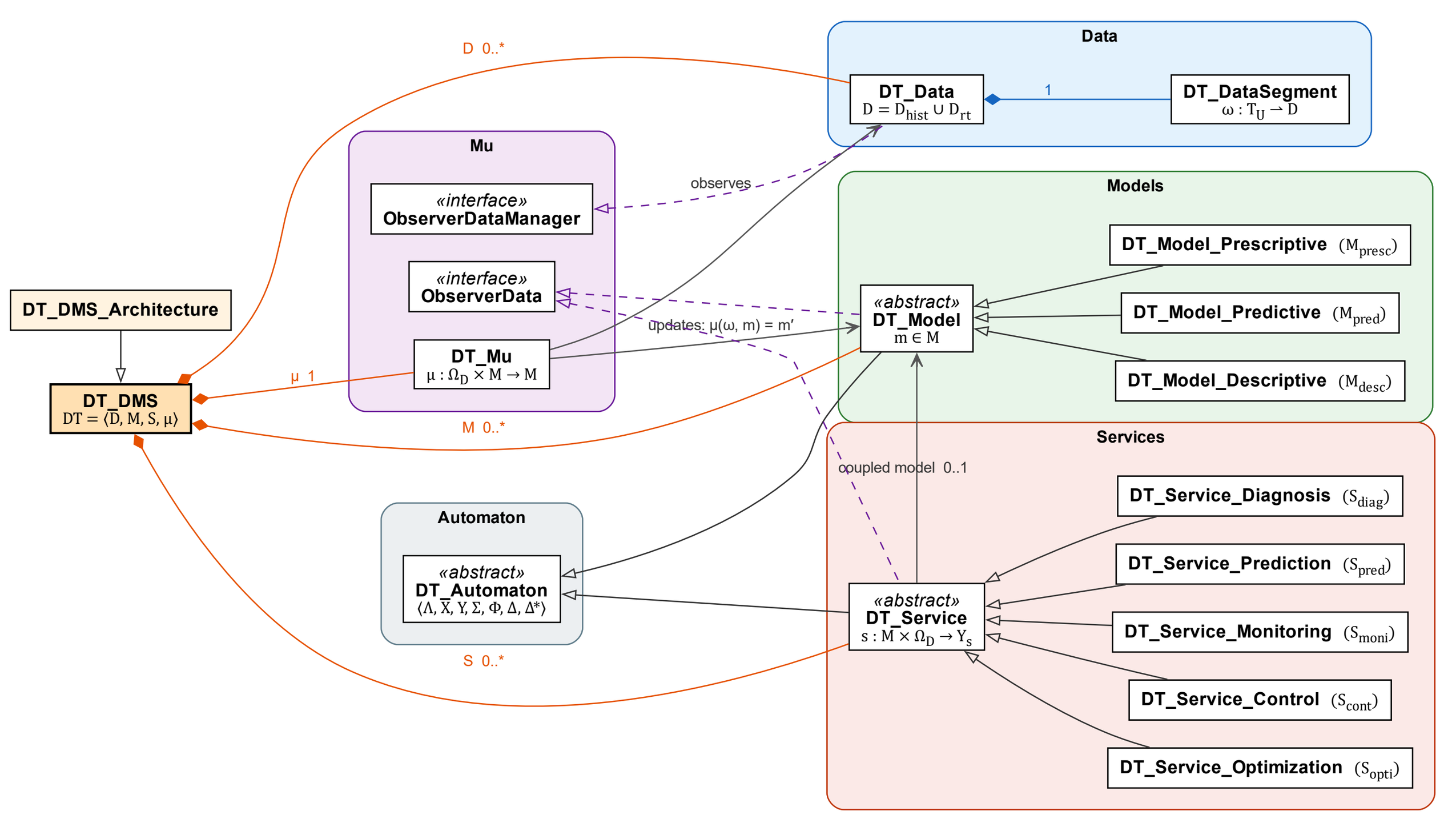

Data
DT_Data
D = D_hist ∪ D_rt
1
DT_DataSegment
ω : T_U ⇀ D
D 0..*
Mu
«interface»
ObserverDataManager
observes
«interface»
ObserverData
DT_Mu
μ : Ω_D × M → M
updates: μ(ω, m) = m′
μ 1
M 0..*
DT_DMS_Architecture
DT_DMS
DT = ⟨D, M, S, μ⟩
Models
«abstract»
DT_Model
m ∈ M
DT_Model_Prescriptive (M_presc)
DT_Model_Predictive (M_pred)
DT_Model_Descriptive (M_desc)
Automaton
«abstract»
DT_Automaton
⟨Λ, X, Y, Σ, Φ, Δ, Δ*⟩
S 0..*
Services
coupled model 0..1
«abstract»
DT_Service
s : M × Ω_D → Y_s
DT_Service_Diagnosis (S_diag)
DT_Service_Prediction (S_pred)
DT_Service_Monitoring (S_moni)
DT_Service_Control (S_cont)
DT_Service_Optimization (S_opti)